

Supporting Information for the publication entitled:

Methane Activation at the Pd/CeO₂ Interface

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1) DFT Methods

DFT total energies are calculated using the VASP 3.3.3¹ with the PW91 exchange-correlation functional.² Following the methodology of our previous study, a $U = 5$ eV Hubbard correction is added to treat spurious electron delocalization in Ce 4f states.^{3, 4} Plane-wave basis sets were truncated at 450 eV, and core electronic regions were represented by the projector augmented wave approximation^{5, 6} (PAW), with the following electronic valence configurations treated self-consistently: $5s^25p^66s^24f^15d^1$ for Ce atoms, $4d^{10}$ for Pd atoms, $2s^22p^4$ for O atoms, $2s^22p^2$ for C atoms, and $1s^1$ for H atoms. Monkhorst-Pack⁷ (MP) Brillouin zone sampling was used with a $2 \times 2 \times 1$ MP k-point spacing, and all calculations were spin polarized. The Pd₃Ce₃₃O₆₉(111) surface was treated with a three Ce-O-Ce tri-layer slab model identical to that employed in our previous publication,⁸ with the addition of an adsorbed Pd₄O₄ cluster as described in the main text. Free energies are calculated as: $G(298\text{ K}) = E_{\text{DFT}}(0\text{ K}) + \text{ZPVE} + \Delta H(0\text{ K} \rightarrow 298\text{ K}) - T\Delta S(0\text{ K} \rightarrow 298\text{ K})$, where standard statistical mechanical expressions for the ideal gas, harmonic oscillator, and rigid rotor are employed for all thermal and entropic contributions to the free energy. Only vibrational contributions are included for calculations involving extended surfaces, as they have no translational or rotational degrees of freedom.

2) Ab initio Thermodynamics

As shown previously,^{8, 9} the relative stability of surface models with varying amounts of oxygen is determined by comparing free energies of formation calculated with the formula:

$$G_{form} = E_{[Pd_7O_x/Ce_{33}O_{66}]} - \left(E_{[Pd_3O_3/Ce_{33}O_{66}]} + 4E_{[PdO]} + \frac{x-7}{2}\mu_{O_2}(T,P) \right) \quad (1)$$

where $E_{[Pd_7O_x/Ce_{33}O_{66}]}$ is the total energy of the surface with an adsorbed Pd_7O_x cluster, $E_{[Pd_3O_3/Ce_{33}O_{66}]}$ is the energy of the $Pd_3O_3/Ce_{33}O_{66}$ surface identified in our previous study (with three incorporated Pd atoms and three oxygen vacancies relative to stoichiometric CeO_2), $E_{[PdO]}$ is the energy of bulk PdO, and $\mu_{O_2}(T,P)$ is the chemical potential of gas phase O_2 . The oxygen chemical potential is calculated as a function of temperature and pressure using standard statistical mechanical formulas. Hence, to determine the thermodynamically stable O:Pd ratio in the embedded cluster, we compare the free energy of surfaces with varying amounts of oxygen, as shown in Figure S1. In the figure, the free energy of each cluster model is plotted against oxygen pressure at 500 K. At each pressure, the thermodynamically stable state is determined by identifying the surface with the lowest free energy. Phase boundaries are located at intersection points where two surface states have equal free energies, indicated by dotted lines in the figure.

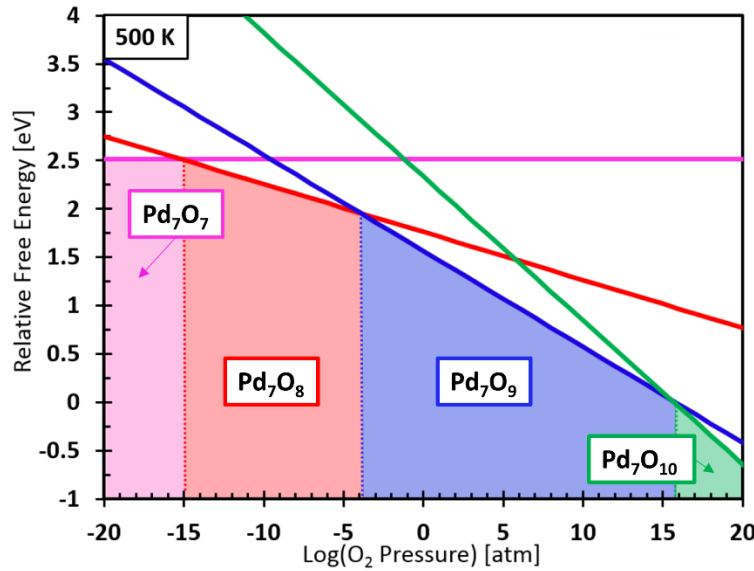


Figure S1. Relative free energy of $Pd_7O_x/Ce_{33}O_{66}(111)$ surfaces with varying Pd oxidation states at 500 K. The vertical dashed lines indicate the pressure regions in which each surface oxidation state is stable.

3) Pseudo-barrier Transition State and Apparent Activation Energy

In this work, we employ a pseudo-barrier approximation to calculate the barrier for methane activation *via* hydrogen abstraction to the oxide surface. In previous publications, this approximation has been validated against explicit transition state searches with the nudged elastic band method. Within this approximation, the transition state is represented by an adsorbed hydrogen atom and a $\cdot CH_3$ radical separated from the surface:

$$\Delta G_{act}(T, P) = E_{[surf-H]} + \mu_{\cdot CH_3}(T, P) - E_{[surf-clean]} - \mu_{CH_4}(T, P) + \Delta E_{ZPV} \quad (2)$$

where $\mu_{\cdot CH_3}$ and μ_{CH_4} are calculated as a function of T and P using standard statistical mechanical expressions, with translational and rotational contributions for $\cdot CH_3$ set to zero (*i.e.*, the $\cdot CH_3$ radical has no translational or rotational degrees of freedom in the activated complex). Additionally, the stability of each cluster model affects the apparent methane activation barrier, and is accounted for in the total apparent barrier expression by including in the barrier expression the state's formation energy relative to the most stable state:

$$\Delta G_{app} = \Delta G_{act} + \Delta G_{form} \quad (3)$$

The formation energy, ΔG_{form} , is calculated as the free energy difference between the surface of interest and the thermodynamically stable surface, which is determined as a function of temperature and oxygen pressure using Equation 1 above. ΔG_{form} is zero for the thermodynamically stable surface at the T, P_{O_2} point under consideration.

4) ReaxFF Parameter Development

Pd/Ce/O cross-terms in the parameter set are optimized to reproduce DFT oxygen vacancy formation energies in the Pd-incorporated surface models reported in our previous study.⁸ Emphasis was placed on obtaining a ReaxFF potential that correctly describes oxygen uptake in Pd-incorporated ceria surfaces, which will allow ReaxFF to reproduce proper $Pd^{\delta+}$ oxidation state transitions as a function of temperature and pressure. This is demonstrated in the phase diagram in Figure S2, which compares DFT and ReaxFF phase boundaries separating adsorbed Pd^0 metal, Pd^{2+} incorporated, and Pd^{4+} incorporated states on the $CeO_2(111)$ surface. As seen in the figure, ReaxFF properly predicts the phase boundary separating the $Pd_3Ce_{33}O_{69}$ and $Pd_3Ce_{33}O_{70}$ surfaces. This boundary reflects the $Pd^{2+} \leftrightarrow Pd^{4+}$ transition responsible for the high activity of the $Pd_xCe_{1-x}O_\delta$ surface, and is therefore the most important boundary for ReaxFF to reproduce. ReaxFF adequately predicts that the $Pd_3Ce_{33}O_{71}$ surface, in which two Pd^{4+} states are present, will not become stable until oxygen pressures well above typical operating conditions are reached. Finally, ReaxFF yields a $Pd_3Ce_{33}O_{68} \leftrightarrow Pd_3Ce_{33}O_{69}$ boundary lying above the $3Pd/CeO_2(111) \leftrightarrow Pd_3Ce_{33}O_{69}$ boundary, indicating that ReaxFF correctly predicts that Pd will egress from the surface to form Pd^0 metal clusters under reducing conditions (*i.e.*, the Pd^0 adsorbed state always has a lower free energy than the Pd^0 incorporated state). ReaxFF, however, significantly under predicts the temperature at which the $3Pd/CeO_2(111) \leftrightarrow Pd_3Ce_{33}O_{69}$ transformation occurs, indicating that ReaxFF simulations will more readily form reduced Pd^0 states compared to DFT predictions. Nevertheless, ReaxFF reproduces the qualitative trends governing $Pd^{\delta+}$ oxidation state transitions, especially regarding the $Pd^{2+} \leftrightarrow Pd^{4+}$ transitions that are the focus of this study.

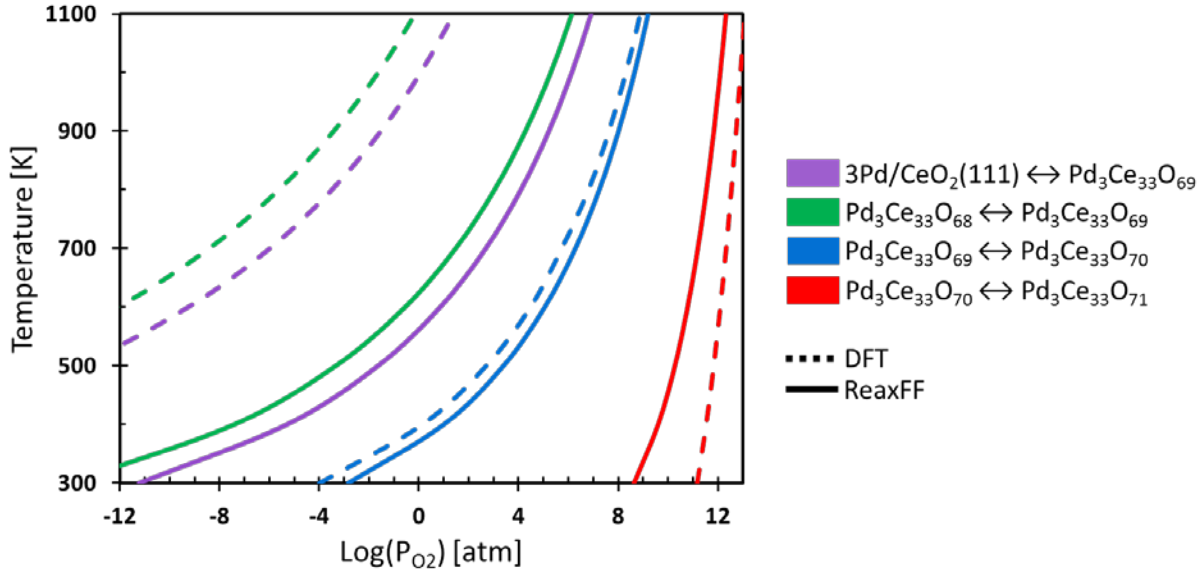


Figure S2. ReaxFF and DFT derived phase diagram describing Pd oxidation state transitions in $\text{Pd}_3\text{Ce}_{33}\text{O}_8(111)$ surface models. The solid data lines represent boundaries derived with ReaxFF, and dashed data lines indicate boundaries derived with DFT.

Various other formation energies and adsorption energies regarding $\text{Pd}/\text{CeO}_2(111)$ surfaces were included in the training set. Formation and adsorption energies were calculated with the following formulas:

$$E_{\text{Surf}}^{\text{Pd-Inc}} = E_{[\text{N}_{\text{Pd-inc}}]} + N_{\text{Pd}}E_{[\text{CeO}_2]} - (E_{[\text{CeO}_2\text{-surf}]} + N_{\text{Pd}}E_{[\text{PdO}_2]} + \Delta N_{\text{O}}\mu_{\text{O}_2}) \quad (4)$$

$$E_{\text{Surf}}^{\text{Pd-Ads}} = E_{[\text{N}_{\text{Pd-surf}}]} + N_{\text{Pd}}\mu_{\text{O}_2} - (E_{[\text{CeO}_2\text{-surf}]} + N_{\text{Pd}}E_{[\text{PdO}_2]}) \quad (5)$$

where N_{Pd} is the number of Pd atoms in the unit cell, ΔN_{O} is the number of oxygens in the system relative to stoichiometric CeO_2 and PdO references, $E_{[\text{N}_{\text{Pd-surf}}]}$ is the DFT energy of the Pd-incorporated or Pd-adsorbed surface, $E_{[\text{CeO}_2]}$ and $E_{[\text{PdO}_2]}$ are the energies of the CeO_2 and PdO_2 bulk references (calculated per stoichiometric formula unit), and $E_{[\text{CeO}_2\text{-surf}]}$ is the energy of the clean CeO_2 surface. The parameterization results are summarized below in Table S1. The Pd-Ce-O bonding, off-diagonal, and angular parameters derived in this study are summarized in Tables S2, S3, and S4, respectively. The remaining C/H/O,^{10, 11} Pd/O,¹² Ce/O,¹³ and Pd/C/H^{14, 15} parameter terms can be found in the provided references.

Table S1. DFT and ReaxFF energies for Pd incorporation and adsorption in ceria surfaces calculated with Eqns. 4 and 5, respectively. All energies are in kcal mol⁻¹.

Pd Incorporation	ReaxFF	DFT
Pd ₁ Ce ₃₅ O ₇₂ (111)	41.98	35.69
Pd ₁ Ce ₃₅ O ₇₁ (111)	47.47	39.51
Pd ₂ Ce ₃₄ O ₇₂ (111)	31.25	35.33
Pd ₂ Ce ₃₄ O ₇₁ (111)	31.67	32.49
Pd ₂ Ce ₃₄ O ₇₀ (111)	35.17	28.48
Pd ₃ Ce ₃₃ O ₇₂ (111)	30.18	34.33
Pd ₃ Ce ₃₃ O ₇₁ (111)	30.53	27.96
Pd ₃ Ce ₃₃ O ₇₀ (111)	30.53	28.07
Pd ₃ Ce ₃₃ O ₆₉ (111)	33.62	31.07
Pd Adsorption		
1Pd + CeO ₂ (111) → 1Pd/CeO ₂ (111)	55.21	67.15
1Pd + CeO ₂ (111)-O _{vac} → 1Pd/CeO ₂ (111)-O _{vac}	142.91	137.74
2Pd + CeO ₂ (111) → 2Pd/CeO ₂ (111)	54.37	63.16
2Pd + CeO ₂ (111)-O _{vac} → 2Pd/CeO ₂ (111)-O _{vac}	95.07	92.61
1Pd + CeO ₂ (110) → 1Pd/CeO ₂ (110)	50.22	58.49
1Pd + CeO ₂ (110)-O _{vac} → 1Pd/CeO ₂ (110)-O _{vac}	79.58	77.53
PdO_x Adsorption		
Pd ₁ O ₁ + CeO ₂ (111) → Pd ₁ O ₁ /CeO ₂ (111)	59.82	58.23
Pd ₁ O ₂ + CeO ₂ (111) → Pd ₁ O ₂ /CeO ₂ (111)	63.77	46.01
Oxygen Adsorption/Desorption		
CeO ₂ + ½O ₂ → O/CeO ₂	22.31	9.64
CeO ₂ → CeO ₂ (111)-O _{vac} + ½O ₂	47.32	56.61
Pd ₁ Ce ₃₅ O ₇₂ (111) → Pd ₁ Ce ₃₅ O ₇₁ (111) + ½O ₂	-5.49	-4.82
Pd ₃ Ce ₃₃ O ₇₀ (111) → Pd ₃ Ce ₃₃ O ₆₉ (111) + ½O ₂	-9.27	-9.98

Table S2. Bond energy and bond order parameters

Atom Pair	D_e (kcal mol⁻¹)	P_{be,1}	P_{be,2}	P_{ovun,1}	P_{bo,1}	P_{bo,2}
Pd-Ce	68.3117	-0.3467	11.8297	0.1073	-0.1205	5.8218

Table S3. Van der Waals interaction and σ-bond parameters

Atom Pair	ε (kcal mol⁻¹)	R_{vdw} (Å)	α_{vdw}	R_{sigma} (Å)
Pd-Ce	0.2452	2.2178	12.8866	2.3042

Table S4. Valance angle parameters

Angle	$\theta_{o,o}$ (degrees)	$P_{val,1}$	$P_{val,2}$	$P_{val,7}$	$P_{val,4}$
O-Pd-Ce	71.0692	9.9527	7.9620	0.1000	1.0102
O-Ce-Pd	1.8709	2.1607	7.9002	0.1000	1.0116
Ce-O-Pd	90.0000	6.7405	7.9834	1.0000	2.1474

5) Charge Density Analysis

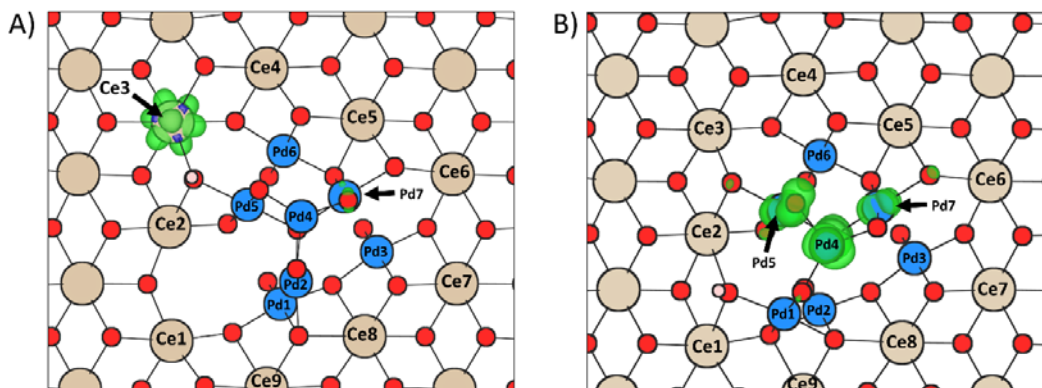


Figure S3. Top view of excess spin density (green) in the (a) Pd_7O_8 and (b) Pd_7O_9 embedded cluster models after the adsorption of a hydrogen atom (white circle) during methane activation (Isosurface = $0.003 \text{ e}^- \text{ bohr}^{-3}$).

Table S5. Bader charge^{16, 17} (in +e) before and after hydrogen adsorption on the Pd_7O_8 model. Bader charge differences with a negative value indicate the accumulation of electron density after hydrogen adsorption.

	Pd_7O_8	H/ Pd_7O_8	Difference
Pd1	0.93	0.93	0.00
Pd2	0.88	0.88	0.00
Pd3	0.94	0.93	-0.01
Pd4	1.22	1.20	-0.02
Pd5	0.95	0.92	-0.03
Pd6	0.99	1.00	0.01
Pd7	0.81	0.82	0.01
Ce1	2.34	2.33	-0.01
Ce2	2.33	2.33	0.00
Ce3	2.36	2.13	-0.23
Ce4	2.38	2.37	-0.01
Ce5	2.36	2.36	0.00
Ce6	2.30	2.30	0.00
Ce7	2.39	2.39	0.00
Ce8	2.32	2.32	0.00
Ce9	2.35	2.36	0.01

Table S6. Bader charge^{16, 17} (in +e) before and after hydrogen adsorption on the Pd₇O₉ model. Bader charge differences with a negative value indicate the accumulation of electron density after hydrogen adsorption.

	Pd ₇ O ₈	H/Pd ₇ O ₉	Difference
Pd1	0.98	0.94	-0.04
Pd2	0.90	0.88	-0.02
Pd3	0.97	0.94	-0.03
Pd4	1.30	1.26	-0.04
Pd5	1.06	1.02	-0.04
Pd6	1.05	1.01	-0.04
Pd7	1.00	0.93	-0.07
Ce1	2.36	2.37	0.01
Ce2	2.36	2.31	-0.05
Ce3	2.35	2.36	0.01
Ce4	2.41	2.41	0.00
Ce5	2.39	2.40	0.01
Ce6	2.34	2.34	0.00
Ce7	2.36	2.36	0.00
Ce8	2.36	2.35	-0.01
Ce9	2.32	2.30	-0.02

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5) Geometry Coordinates and Total DFT Energies

Pd₇O₇/Ce₃₃O₆₆(111)

Total DFT Energy: -869.45905821 eV

Unit Cell (Å):

13.3200000000000003	0.0000000000000000	0.0000000000000000
0.0000000000000000	11.3610000000000007	0.0000000000000000
0.0000000000000000	0.0000000000000000	30.0000000000000000

Fractional Coordinates:

Pd	0.5225266115834529	0.4937280673227326	0.3268664855894126
Pd	0.2715794605483838	0.8119081876579883	0.3276861866640391
Pd	0.2647496041969107	0.3305308793806892	0.3291504608740489
Pd	0.4352309141915892	0.6643296459237772	0.3877476401992591
Pd	0.1702112212625855	0.6447186147404544	0.3880274341318382
Pd	0.3086777223490574	0.5877863600004042	0.4694058722110265
Pd	0.3058035236097458	0.4208256966601366	0.4093256666050644
Ce	0.9730204013295195	0.2152816322734582	0.3464638444666094
Ce	0.7361188746867858	0.3963990203246961	0.3491541987063378
Ce	0.4887310717463327	0.2308983815025401	0.3484023565161354
Ce	0.7242661968311757	0.7362071273952541	0.3506438190070985
Ce	0.9778497099042732	0.8948752179520099	0.3512709477522370
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Ce	0.4011324545887022	0.3983828274772180	0.2391482213848606
Ce	0.6524345236400361	0.5665928758619114	0.2396079591744275
Ce	0.6501649700410046	0.2309612973293224	0.2401765477559271
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Ce	0.6477054195016203	0.8984635791432362	0.2398436907201574
Ce	0.1465865718201632	0.9003690859623502	0.2421416566790269
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Ce	0.3157315851271812	0.8990392902604926	0.1356679461433540
Ce	0.8162925995492959	0.8983055936052597	0.1322826639184241
Ce	0.3160333035648908	0.2316439069883219	0.1347226536928560
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Ce	0.8169406746968491	0.5657045178806531	0.1329101455601801
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Ce	0.0663495155965946	0.0648115123831957	0.1336300877775505
Ce	0.3144657781267414	0.5663409949502145	0.1364863695307826

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O	0.6517794680253829	0.2287499754828846	0.3196785739930635
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O	0.6472112976789149	0.9009105491861754	0.3201708083605819
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O	0.9809375786105116	0.8991437704449844	0.2679894161471130
O	0.9735328437609946	0.5655444707107904	0.2681657343738493
O	0.0643150824561560	0.3985387813503988	0.2155792303417626
O	0.3144506313881804	0.2324212246859841	0.2138642832485869
O	0.3152420345936735	0.5649833038549348	0.2153951445827534
O	0.2294851678366245	0.3985778173947437	0.2689062449803054
O	0.4821066126239882	0.9015547715881417	0.2688327935900441
O	0.3139029261393027	0.9016535547336859	0.2148490110964354

O	0.0656297993018337	0.7313515952077007	0.2156208931115204
O	0.2340509465465455	0.7317443938976522	0.2692470639167222
O	0.9854333375038370	0.8987108017137458	0.1066011566396466
O	0.4816741910937853	0.8981900497409715	0.1071861585392864
O	0.6496321721214460	0.8978378668605249	0.1599060455198306
O	0.1491645217729649	0.8982092523031411	0.1625318239262995
O	0.7342636046050012	0.0648683635919868	0.1057051135742608
O	0.8997303162882214	0.0647647776426311	0.1603285623724688
O	0.4828563780912744	0.5661714848752057	0.1076072936772726
O	0.7349584093835457	0.3986408356299120	0.1059576096914477
O	0.7352831238461860	0.7314618483959099	0.1056569891724877
O	0.4831450453239954	0.2322188109165804	0.1069036677187848
O	0.4010772896024014	0.3986247837982779	0.1618940475427186
O	0.6505295146856366	0.2313795146450606	0.1601831710170184
O	0.6505070288018583	0.5660453845791105	0.1601310034956092
O	0.3995470600547904	0.0660849029132839	0.1619369540746164
O	0.9007622404057932	0.7319551780552098	0.1608553384033514
O	0.9004587023208205	0.3987955533021498	0.1615091250708730
O	0.9849648901615567	0.2323754251811341	0.1069807778783662
O	0.2333328524236836	0.0653731991384228	0.1076472699228363
O	0.2338218285998936	0.3992528588931643	0.1084801472575655
O	0.1493839030969163	0.2324391932598074	0.1624950076645151
O	0.3989411624981173	0.7329527606205885	0.1630409493308085
O	0.2334131449920829	0.7309080840230228	0.1084502336539738
O	0.9860091480072941	0.5649570903301600	0.1071643150019527
O	0.1492414255189119	0.5649282295195781	0.1639747052998560
O	0.4381915936821012	0.6351243106881578	0.4522969841496846
O	0.2004941701509581	0.6829938792927738	0.4501864731509675
O	0.3048216108554948	0.5713502349662577	0.3779529638985089
O	0.0322939754413866	0.7321903742747109	0.3863749646133444

*H** on $Pd_7O_7/Ce_{33}O_{66}(111)$

Total DFT Energy: -874.23366834 eV

Unit Cell (Å):

13.320000000000000	0.000000000000000	0.000000000000000
0.000000000000000	11.361000000000007	0.000000000000000
0.000000000000000	0.000000000000000	30.000000000000000

Fractional Coordinates:

H	0.5107199653202765	0.6468542151336506	0.4654566427283364
Pd	0.5218832521244908	0.4924984905608167	0.3266340933261501
Pd	0.2695013628721037	0.8119297669498317	0.3287648017944650
Pd	0.2634324932331659	0.3312780732981630	0.3295738794783253
Pd	0.4373230769317907	0.6581380203855738	0.3867726568396971
Pd	0.1688748643448852	0.6421192701973141	0.3873179683731542
Pd	0.3182341955008503	0.6062200303850276	0.4640727715424168
Pd	0.3043878079938100	0.4168889743608715	0.4126325229155050

Ce	0.9724872226493418	0.2154730796706174	0.3469200070187416
Ce	0.7352466607875136	0.3964423568123120	0.3493599498321358
Ce	0.4868141170279706	0.2310749993530259	0.3490298230085389
Ce	0.7241040939465497	0.7362321167890102	0.3505693109957508
Ce	0.9770855665083183	0.8951732425253793	0.3516385641317458
Ce	0.9683299156224157	0.5790176494652527	0.3468081358041089
Ce	0.7291402471714092	0.0634744390319720	0.3484520534334088
Ce	0.4829840776190993	0.9082511366586586	0.3503545930421282
Ce	0.2280989926562596	0.0688183706342236	0.3510993669944290
Ce	0.3977094604572260	0.0668181967367180	0.2426100359964916
Ce	0.8967435791633515	0.0631681873302395	0.2396192285780273
Ce	0.4003523816724274	0.3985503213789101	0.2393985477051504
Ce	0.6514036305081140	0.5670977387286271	0.2398116427550440
Ce	0.6493517994686157	0.2314782952311576	0.2409389959345105
Ce	0.8964949161627386	0.7346884955007191	0.2400945095056370
Ce	0.8984186249867193	0.3982749450767500	0.2405977898620863
Ce	0.1458745916879589	0.2323166325696282	0.2414565972996412
Ce	0.3991478820572981	0.7329659379366831	0.2427200334093411
Ce	0.6470488185315866	0.8988193144616664	0.2403733888783077
Ce	0.1460158715661736	0.9009843988071710	0.2424071151834681
Ce	0.1459032657293473	0.5677591966411187	0.2475689909407144
Ce	0.3154433427215013	0.8998514814331843	0.1360231459378514
Ce	0.8158148161163747	0.8990291286630995	0.1327436413112421
Ce	0.3155427909058692	0.2321302890931502	0.1350212676499348
Ce	0.5668148424123037	0.3993605700505307	0.1337451385967111
Ce	0.5655873638750117	0.0657776407737526	0.1343030691866220
Ce	0.8164716827120779	0.5663642395812940	0.1333453687051449
Ce	0.8162792615678873	0.2324811590245474	0.1335010257853757
Ce	0.0658324193325737	0.0654597087727778	0.1342346899309854
Ce	0.3141291977787811	0.5669826168816811	0.1365418037195553
Ce	0.5646138809203851	0.7326989444385559	0.1340794465557086
Ce	0.0673431565824079	0.7315002251223748	0.1357876775440424
Ce	0.0669851189982110	0.4004685637093107	0.1356502456463467
O	0.8131057711646219	0.2263552817341221	0.3768541605232953
O	0.6507678970376971	0.2283585058217358	0.3201982487886382
O	0.4020569096408539	0.3895331660225820	0.3181289739366315
O	0.6544517852028487	0.5696935234855751	0.3207660704097223
O	0.5682905686996977	0.7570132780158982	0.3846385408332670
O	0.8097314727394535	0.5676226131961829	0.3763275992155395
O	0.8084844378899663	0.8995291953305758	0.3775709982283277
O	0.5764412996422287	0.3892130662726041	0.3774273721018995
O	0.8906795399673936	0.7382569046221317	0.3203074218386829
O	0.9094293970936643	0.3943709737153078	0.3277480020043294
O	0.0553185936233840	0.0638146364675165	0.3780219461808494
O	0.1352278593692975	0.2400528269348572	0.3273707446118386
O	0.3152831042055300	0.2345901261480577	0.3836550567796875

O 0.6459920226711819 0.9009200370343374 0.3207962387945853
 O 0.3979664929683027 0.0742395605380219 0.3218596911390182
 O 0.4047267188561313 0.7339978389153570 0.3270288674642403
 O 0.5658776761843914 0.0697526194059786 0.3776449785475443
 O 0.8938153144312924 0.0597633829616101 0.3208191359900583
 O 0.1252380303932040 0.5782121433443753 0.3246692448071556
 O 0.1405238115747665 0.9037946021580233 0.3230634319506292
 O 0.3166588494451901 0.9195826512635711 0.3794473028010066
 O 0.0641447132663915 0.0659999629275685 0.2136314458261843
 O 0.5651642840871697 0.0642140486641274 0.2131693271482625
 O 0.7321046728775343 0.0640434033551259 0.2676448822315421
 O 0.2272930938401220 0.0712478981233443 0.2713034422109105
 O 0.8147346600100095 0.2322650416540614 0.2126010510862285
 O 0.9779908113752050 0.2278141791259007 0.2689980899174012
 O 0.5648314950602195 0.7329247148685518 0.2130684881961025
 O 0.8137528579713412 0.5659544276612797 0.2122909287507195
 O 0.8137786644387982 0.8990456027524848 0.2120619037519569
 O 0.5686381150620783 0.3996112955744532 0.2129320363124513
 O 0.4839375743081511 0.5683286746948587 0.2671310231288380
 O 0.7387087841235459 0.3985514139676708 0.2702646692789396
 O 0.7301784453905630 0.7341687786403087 0.2677398957460376
 O 0.4856420777116497 0.2302730258984295 0.2669822521548840
 O 0.9800097965842206 0.8997077133192847 0.2683024134606923
 O 0.9726195232679775 0.5661870044620115 0.2683688500136031
 O 0.0637368952558538 0.3990402986464000 0.2158176494508984
 O 0.3137944172202067 0.2327449203878061 0.2141523214823588
 O 0.3144342186718310 0.5657466803021570 0.2155186435875745
 O 0.2290364972964070 0.3994532506333972 0.2691129174775257
 O 0.4808032487854863 0.8999449874940021 0.2695641427652961
 O 0.3135452088854659 0.9020094219820772 0.2151009762183827
 O 0.0650361215767321 0.7322736179056882 0.2158278239261813
 O 0.2337225526425318 0.7326869268399531 0.2697019724129039
 O 0.9851283882122354 0.8992947930303149 0.1070266560990756
 O 0.4814350684913165 0.8988896116904369 0.1076341107419934
 O 0.6492273660107275 0.8986047770123907 0.1603805763982162
 O 0.1488361144790825 0.8989903019655503 0.1628890961388405
 O 0.7338104841365222 0.0657278951783980 0.1062422549502894
 O 0.8991929941149613 0.0652078983399796 0.1607867627955868
 O 0.4822036174284342 0.5670669682093187 0.1078940262949933
 O 0.7342327566862457 0.3991448314077423 0.1064739976470641
 O 0.7347581474794124 0.7324216837372265 0.1061090400373732
 O 0.4826874758071973 0.2330494267906659 0.1072362392230644
 O 0.4003708095278487 0.3993146292380254 0.1621590280114813
 O 0.6495633527582336 0.2318409271567468 0.1607576780887982
 O 0.6498743309445246 0.5667194898585466 0.1604405581736482
 O 0.3992975174234232 0.0667036152861924 0.1622960523885051

O	0.9002346278057755	0.7327124738046588	0.1612248486295219
O	0.8999058979040025	0.3995337357469635	0.1619156628661026
O	0.9841412883509234	0.2328936370615948	0.1074401705574454
O	0.2330485867675768	0.0660450401358761	0.1080180187630352
O	0.2333132941874907	0.3998799879520533	0.1086249609316029
O	0.1488507521823758	0.2330326893906963	0.1628153785100833
O	0.3984387459106746	0.7335583305857730	0.1634686779543865
O	0.2332585791797973	0.7319306493457730	0.1086823852035116
O	0.9852675290781784	0.5655901117693535	0.1075016605080451
O	0.1487290683142467	0.5657050509785200	0.1640420240352387
O	0.4694861004818346	0.5878119202922016	0.4504786423970838
O	0.1938763845354645	0.6770237149858777	0.4504526700961008
O	0.3065384147213208	0.5641767720329233	0.3785220919261778
O	0.0322879697632711	0.7319794248777942	0.3861537205550720

Pd₇Og/Ce₃₃O₆₆(111)

Total DFT Energy: -875.62069302 eV

Unit Cell (Å):

13.3200000000000003	0.0000000000000000	0.0000000000000000
0.0000000000000000	11.3610000000000007	0.0000000000000000
0.0000000000000000	0.0000000000000000	30.0000000000000000

Fractional Coordinates:

Pd	0.5219885983853181	0.4934393236366568	0.3271530677329458
Pd	0.2710216861063452	0.8098792239889149	0.3272918488155972
Pd	0.2644340414137459	0.3307837308122287	0.3297190716234034
Pd	0.4331099974972514	0.6650507236410502	0.3893357476648751
Pd	0.1736291861094177	0.6471282143100694	0.3881820943862280
Pd	0.3106652588843632	0.6192317515438672	0.4738991278484134
Pd	0.3009325886201671	0.4026085664359147	0.4121952678836963
Ce	0.9730110135152067	0.2156175279842752	0.3464714135210278
Ce	0.7361616362736348	0.3964749330831640	0.3491863614037259
Ce	0.4897427084330896	0.2291673731702157	0.3480482030308726
Ce	0.7244530625664823	0.7357925979549305	0.3503305686760316
Ce	0.9776214021812368	0.8952027668009892	0.3508968020823867
Ce	0.9697896130435876	0.5790974091094603	0.3473701767023774
Ce	0.7301495022660627	0.0626361449464560	0.3476649399432594
Ce	0.4836661742243293	0.9054695207937346	0.3491189713417646
Ce	0.2279216262476113	0.0662158113053750	0.3504210193963226
Ce	0.3982767365730209	0.0658899230482918	0.2419673641411980
Ce	0.8971612580459175	0.0626062125460706	0.2390185921194372
Ce	0.4014564466366473	0.3977072830341614	0.2391445832573872
Ce	0.6523427585927380	0.5663344897445513	0.2395735594991022
Ce	0.6502631842696106	0.2308451168924298	0.2399160496479137
Ce	0.8971311621762023	0.7337187131233984	0.2398961636961174
Ce	0.8990604485738786	0.3979228890708720	0.2405734893632352
Ce	0.1462795246032647	0.2309012352624942	0.2406901342080052

Ce	0.3998809347801108	0.7317901581339604	0.2424101509703736
Ce	0.6477606809019809	0.8980135751656489	0.2395951065411990
Ce	0.1463983795606932	0.8998792362779321	0.2417437461983011
Ce	0.1465132394282448	0.5660934391181760	0.2477248507681577
Ce	0.3155570531080912	0.8987031384833637	0.1354101170869287
Ce	0.8159927185523365	0.8984729697333723	0.1322384532557306
Ce	0.3159434133780723	0.2316307944953624	0.1344584912031489
Ce	0.5674257566128726	0.3990515742423894	0.1331293053867437
Ce	0.5656613620080011	0.0652751170416401	0.1334427374697298
Ce	0.8169304516050275	0.5657997902118501	0.1329649656917089
Ce	0.8167594390294379	0.2319905205291937	0.1329162717169548
Ce	0.0660762040139592	0.0646489666418543	0.1334508995624560
Ce	0.3142832265966452	0.5659908049405796	0.1362177550722612
Ce	0.5648860899130367	0.7318142501878664	0.1334115040132482
Ce	0.0674105373793565	0.7305616338571798	0.1354351013605234
Ce	0.0672458030076097	0.4000319225622093	0.1354110856268199
O	0.8143058240876645	0.2261971093895241	0.3762397201779670
O	0.6519804370598723	0.2284054385499063	0.3194828362160648
O	0.4030643096686963	0.3889648380589970	0.3187169599365832
O	0.6551097019660587	0.5690835384067487	0.3206322568843938
O	0.5679511093893633	0.7574614968508379	0.3824991062404676
O	0.8107761159180239	0.5669636545049122	0.3762342555171702
O	0.8096212756144748	0.8995728421471201	0.3767908101764293
O	0.5772449426608990	0.3886236585933767	0.3774036701619848
O	0.8915165780550410	0.7371823643327786	0.3201221024889919
O	0.9100717041583075	0.3942419555458092	0.3272983412713850
O	0.0563503539383892	0.0630579533004345	0.3771338786007348
O	0.1365194427936480	0.2394001808082128	0.3269955365235323
O	0.3146877091632753	0.2314432625629606	0.3833673181370074
O	0.6474666713466372	0.9000480000406638	0.3198093325103688
O	0.3985142347299240	0.0741709893485014	0.3211266141428767
O	0.4057311390439020	0.7293627475816724	0.3244634921967963
O	0.5673974854568107	0.0687712036905080	0.3763494222526768
O	0.8945215029461147	0.0600694331324683	0.3200675602687725
O	0.1265915403642993	0.5728779451299100	0.3268690508325126
O	0.1415433089707743	0.9018930359628369	0.3223289491321037
O	0.3177861252017637	0.9177853741471141	0.3778501709786880
O	0.0645790447115218	0.0651133433647260	0.2129984683092874
O	0.5656489125260363	0.0641823570438140	0.2122850145694093
O	0.7327738311942734	0.0634259674137600	0.2668106036580861
O	0.2274263390176366	0.0704496734382689	0.2707636623199799
O	0.8155035052141366	0.2319968735802257	0.2122287940789735
O	0.9790442316187585	0.2272715801742681	0.2685113178064117
O	0.5655320760182047	0.7318311313515647	0.2125219379428738
O	0.8148726748609203	0.5651573218263122	0.2121475756573378
O	0.8143471872479912	0.8981110855713534	0.2114288621683132

O 0.5696349027569577 0.3988137969460474 0.2123355811015313
 O 0.4851623527008323 0.5663398189338158 0.2667381567977621
 O 0.7394344893056979 0.3983371506082370 0.2698447623135906
 O 0.7311332897439309 0.7331975351435409 0.2672066404061754
 O 0.4866063079441357 0.2300447766325799 0.2665507130001788
 O 0.9804946133935992 0.8991196506079107 0.2678168923247840
 O 0.9749191787039296 0.5655271870414520 0.2685427228426685
 O 0.0648628434509266 0.3984469859266346 0.2157510857880934
 O 0.3144532455857377 0.2319721933017979 0.2137279448847180
 O 0.3151297152424051 0.5637054136969720 0.2152042466254157
 O 0.2307217783039983 0.3976483752254140 0.2691759486283622
 O 0.4819368027465368 0.8999407666806822 0.2685394474002815
 O 0.3137298139006303 0.9009901470952490 0.2146429058117314
 O 0.0654313025146751 0.7309711496533792 0.2155435860982536
 O 0.2331670574674820 0.7301703587773486 0.2689242828611917
 O 0.9852064325532363 0.8987693283484911 0.1065808640884544
 O 0.4813899059987605 0.8980169109451588 0.1070057182846063
 O 0.6493538674112260 0.8980760000896969 0.1597059892420954
 O 0.1490200880660858 0.8979278373780168 0.1622996759031057
 O 0.7342355267168311 0.0653140197082061 0.1056206935802426
 O 0.8994596733060517 0.0649206796722674 0.1602613957864066
 O 0.4829541094062035 0.5660978129804228 0.1073384405503136
 O 0.7351579367034718 0.3988266917556907 0.1059448223520575
 O 0.7350988156967261 0.7317677390279459 0.1056490284212575
 O 0.4831754036797957 0.2322663521792767 0.1066162511167573
 O 0.4011442307284654 0.3983428320246298 0.1616655921972308
 O 0.6503677885090897 0.2314126801129823 0.1600518332148506
 O 0.6506453999861121 0.5659889871885011 0.1600099530313437
 O 0.3993674446112978 0.0659291422900534 0.1616556449789586
 O 0.9006217349478196 0.7319115586557506 0.1608335012946351
 O 0.9007260856771557 0.3989616279329201 0.1616888952582860
 O 0.9847447451083211 0.2324730631305482 0.1069217040872937
 O 0.2330936470907050 0.0652190249577084 0.1074577867670705
 O 0.2335857475408720 0.3990029633505989 0.1082475436396515
 O 0.1492775867355608 0.2323473903060775 0.1623936068660695
 O 0.3987729032503423 0.7324313023619536 0.1628493732107986
 O 0.2333151658873930 0.7307134845552369 0.1082619228929652
 O 0.9856662366091293 0.5649603654786614 0.1071227037722139
 O 0.1491474940025031 0.5647531320851816 0.1638322240309958
 O 0.2929574465811343 0.4548349287137395 0.4735312987178658
 O 0.4400095470087519 0.6582763570139447 0.4544638839555896
 O 0.2029152946494609 0.7037242565141770 0.4488103224764358
 O 0.3055831074423637 0.5622422667736310 0.3829259401621795
 O 0.0353447202756230 0.7341802284857513 0.3850656921677530

H on Pd₇O₈/Ce₃₃O₆₆(111)*

Total DFT Energy: -880.35251573 eV

Unit Cell (Å):

13.320000000000000	0.000000000000000	0.000000000000000
0.000000000000000	11.361000000000007	0.000000000000000
0.000000000000000	0.000000000000000	30.000000000000000

Fractional Coordinates:

H	0.0274160551950306	0.7268639147157449	0.4327249154372799
Pd	0.5215288131872809	0.4943214466592560	0.3268135336769460
Pd	0.2721109263804675	0.8097273623426058	0.3274395172425246
Pd	0.2635678272412785	0.3314121873451706	0.3282673654579300
Pd	0.4356758726358283	0.6669294335201594	0.3898299413562872
Pd	0.1764052546169001	0.6413464428808038	0.3919876337248659
Pd	0.3200445595721410	0.6052831113948274	0.4756158900185555
Pd	0.3054655823043123	0.4012970966102378	0.4096556059558192
Ce	0.9733517511516220	0.2163857394065289	0.3459665122815633
Ce	0.7348229900903346	0.3953385709156195	0.3489789204072291
Ce	0.4889401331809197	0.2303975007653817	0.3473545174538462
Ce	0.7239758114526328	0.7375906414405644	0.3502470470700626
Ce	0.9766600017446705	0.8996928594229336	0.3521462389787682
Ce	0.9673729953736685	0.5729145670212067	0.3465499557379763
Ce	0.7288693178032938	0.0635682499929304	0.3475337443848147
Ce	0.4831712191642171	0.9075298107645692	0.3490228036572521
Ce	0.2269848810209622	0.0672967438394582	0.3495851560787943
Ce	0.3978383640147830	0.0664990792712214	0.2415053290126774
Ce	0.8967410272608675	0.0628146610919311	0.2387393298128721
Ce	0.4012341979643592	0.3985254632659391	0.2383832560170826
Ce	0.6525703841219744	0.5668449842552725	0.2395594955599773
Ce	0.6498430924107984	0.2314394793049522	0.2393243278666394
Ce	0.8973894885584914	0.7343395819545363	0.2404277081992243
Ce	0.8988034820848111	0.3977991305119994	0.2395308247995877
Ce	0.1456122675536284	0.2314125272403848	0.2396628946855836
Ce	0.3999078689390816	0.7323904208807763	0.2424190290508179
Ce	0.6473357720674972	0.8990562488168146	0.2395019420197499
Ce	0.1462744296742176	0.9003721865857788	0.2417387472060079
Ce	0.1465697845520407	0.5673618449313865	0.2458445982396403
Ce	0.3154181590816085	0.8987589865794812	0.1352714623867122
Ce	0.8159085774250504	0.8985937259897860	0.1322675827576133
Ce	0.3158485998083828	0.2318444550866045	0.1338601168093607
Ce	0.5675358795402414	0.3993556151507180	0.1327766290644927
Ce	0.5652464279251274	0.0652226824139514	0.1330469890175012
Ce	0.8169609997861119	0.5664661220197335	0.1328459583856540
Ce	0.8165532512580140	0.2321280854833476	0.1324329185878958
Ce	0.0659600850685445	0.0645137097700861	0.1330617379674864
Ce	0.3148245537085028	0.5666776611155596	0.1356220025118877
Ce	0.5650534468634255	0.7322445455882236	0.1335418752219849

Ce 0.0671752492738322 0.7309857703226736 0.1351952488675695
 Ce 0.0672169848421418 0.4002217032716697 0.1345446820815870
 O 0.8128822589753777 0.2279343629937929 0.3764564415870463
 O 0.6510414820488185 0.2291115555783794 0.3189480029577336
 O 0.4023037178587174 0.3902144652031047 0.3180134256562017
 O 0.6550666827401619 0.5694613715172503 0.3207957779722615
 O 0.5689719491405845 0.7599706911444816 0.3830864965604084
 O 0.8122307477574653 0.5636532521905498 0.3765639795491301
 O 0.8007448786637582 0.8998546350276211 0.3776278584982186
 O 0.5759442166375195 0.3890860275278043 0.3771806491681729
 O 0.8932926121504453 0.7322102520589294 0.3212149196430466
 O 0.9090291793120036 0.3937475906918208 0.3261639632602009
 O 0.0617511474454609 0.0751179069401137 0.3779925408467827
 O 0.1358663717177813 0.2412866988539636 0.3233868218042800
 O 0.3125602348396137 0.2316294949831850 0.3821663758060079
 O 0.6455278662957152 0.9011636151539062 0.3190489687279060
 O 0.3982038810266867 0.0752508761096398 0.3206150631719475
 O 0.4073874230338949 0.7299747467349081 0.3251997323932679
 O 0.5660777012441677 0.0708026796326189 0.3762846153636046
 O 0.8929532513619013 0.0637239133737526 0.3198865999288647
 O 0.1229597407147465 0.5814299143609559 0.3293203210553449
 O 0.1434861772099398 0.9020514890779160 0.3222757939952295
 O 0.3177386800396348 0.9187211336228003 0.3777739547356765
 O 0.0648090485861998 0.0656584237255293 0.2123754754461084
 O 0.5649773361926625 0.0647867070216966 0.2118981180998271
 O 0.7320622268684732 0.0641904674291962 0.2662394626671881
 O 0.2277050527303653 0.0706337977050275 0.2698145973611668
 O 0.8148575652218384 0.2323826639145899 0.2113565191791140
 O 0.9782050727319638 0.2279919810109474 0.2672678821324878
 O 0.5656174663320623 0.7323863150937089 0.2126559145547985
 O 0.8150255788717202 0.5657073739897365 0.2121015851427848
 O 0.8135844168364643 0.8979858316716441 0.2111974744515241
 O 0.5692354830266743 0.3995079853383482 0.2119138577304572
 O 0.4850597673462791 0.5669113141152251 0.2663979803355216
 O 0.7388996100492262 0.3986713372548447 0.2693020337552841
 O 0.7317637205606742 0.7339668363095012 0.2674843682018601
 O 0.4859722037699785 0.2310416819186221 0.2659884554338548
 O 0.9796420709494583 0.8997270180226530 0.2664614727059879
 O 0.9763916712520844 0.5639880273056128 0.2679663447711392
 O 0.0648888539277086 0.3988715713403254 0.2146400680271015
 O 0.3139871374744619 0.2332733213193162 0.2130908239600853
 O 0.3152190673977859 0.5647977992502731 0.2144538675422039
 O 0.2299866255827519 0.4007355100511827 0.2684456728281777
 O 0.4806821758696965 0.9007696921674840 0.2681844198297285
 O 0.3134416774408373 0.9007620602840427 0.2143778411699156
 O 0.0658340507907177 0.7307020478671939 0.2153442626126432

O	0.2335390857474136	0.7294907407057061	0.2690128454949621
O	0.9852846920118635	0.8986856332000507	0.1062984484660955
O	0.4815443046299974	0.8982056471090158	0.1067532037071342
O	0.6491035857055805	0.8983862039265391	0.1595522516085544
O	0.1488764138521622	0.8976933218592300	0.1620518434936991
O	0.7338557128085483	0.0652541940214248	0.1053381816148234
O	0.8993384706936551	0.0648864375194509	0.1598701222871816
O	0.4831666593943178	0.5666103207525226	0.1071063926528765
O	0.7347088273495298	0.3993025836639161	0.1055944391134937
O	0.7353804057621584	0.7318766075374744	0.1055491555918693
O	0.4827838501997178	0.2323526065033030	0.1062210557915925
O	0.4011613128581544	0.3986833179913896	0.1610883711102792
O	0.6499290383200140	0.2317425075030104	0.1596272998985715
O	0.6508462089327837	0.5665012971620657	0.1600384175920562
O	0.3987525585276638	0.0661580950337515	0.1612875587720417
O	0.9009048606304461	0.7320947764296403	0.1609027237130775
O	0.9005647203669461	0.3994382684925816	0.1609718219954924
O	0.9843293717049887	0.2327172659316003	0.1064076802810418
O	0.2330093620419893	0.0648825908968770	0.1071257985427103
O	0.2335004354137717	0.3999567574996755	0.1077228520139001
O	0.1490454425499468	0.2324521480499813	0.1613175968767641
O	0.3991652397389223	0.7329565755159705	0.1626603824063151
O	0.2335751443168963	0.7307263895103685	0.1079462788139423
O	0.9858789674604167	0.5656739853840429	0.1066785625338180
O	0.1491844857120141	0.5652505539278699	0.1636452930648041
O	0.3073058803197173	0.4407940076208400	0.4725468717942036
O	0.4452220481066336	0.6553878952308047	0.4548736463813670
O	0.2071949386249077	0.6880549912292225	0.4536564622214800
O	0.3090857346402716	0.5639387407470386	0.3829035109039039
O	0.0323377956549967	0.7217347522196135	0.4003261455915187

Pd₇O₉/Ce₃₃O₆₆(111)

Total DFT Energy: -881.22262826 eV

Unit Cell (Å):

13.3200000000000003	0.0000000000000000	0.0000000000000000
0.0000000000000000	11.3610000000000007	0.0000000000000000
0.0000000000000000	0.0000000000000000	30.0000000000000000

Fractional Coordinates:

Pd	0.5241949886840545	0.4994641083348418	0.3268713723607671
Pd	0.2753524610761861	0.8139305280160561	0.3274181026800553
Pd	0.2735192818720259	0.3349172716559194	0.3272452409896907
Pd	0.4339407475156795	0.6552899486093156	0.3854856857729437
Pd	0.1795458092682135	0.6484889457976885	0.3832986143734098
Pd	0.2982576485824380	0.5342217777677346	0.4549294285362646
Pd	0.1722170464275059	0.3295362765855204	0.4020050200101236
Ce	0.9734154462954450	0.2311755364109439	0.3531841723868675

Ce	0.7364372293900915	0.4008868758038537	0.3494185449753838
Ce	0.4928154035980046	0.2329804819127362	0.3488987139323745
Ce	0.7299008679824074	0.7377858780208081	0.3513163767997196
Ce	0.9813696022998860	0.8963015691726094	0.3514660304911025
Ce	0.9751084562589079	0.5663186820975540	0.3516863060585930
Ce	0.7346541508357529	0.0666445031619676	0.3492005212977783
Ce	0.4878465598895599	0.9101843844594056	0.3506477890882235
Ce	0.2290643427707167	0.0677335819746850	0.3520477125315969
Ce	0.4016939169715021	0.0712926692499561	0.2427444821344781
Ce	0.9021857220390884	0.0684751510088530	0.2427381587966635
Ce	0.4042035126834746	0.4024456895026000	0.2375001406291494
Ce	0.6551328168775976	0.5705698180523270	0.2396452871376699
Ce	0.6533295720198518	0.2357014092912936	0.2409111472505589
Ce	0.9005500482540604	0.7359805463093574	0.2428852593642107
Ce	0.9019651909016589	0.4012729413267658	0.2422049868351457
Ce	0.1495067647890480	0.2369310520249554	0.2392809058466825
Ce	0.4041505970198058	0.7374506283283431	0.2417110320543363
Ce	0.6525393015122889	0.9031175057531230	0.2414416094360840
Ce	0.1501091255905735	0.9055448129277550	0.2423514932659589
Ce	0.1491667170903487	0.5719210055764838	0.2470979691740554
Ce	0.3194523317260523	0.9044150383046491	0.1356024363447206
Ce	0.8203693457187109	0.9036117301325667	0.1348592263273355
Ce	0.3196651983885531	0.2357150936360835	0.1337625484263623
Ce	0.5714714565590153	0.4029902870074501	0.1331988896972506
Ce	0.5702150566862486	0.0692505144011391	0.1346894702249717
Ce	0.8207145447592812	0.5708957255860578	0.1343768437201477
Ce	0.8207791779534321	0.2364490602540885	0.1344369160962235
Ce	0.0686542520125061	0.0689702892841396	0.1343392912181829
Ce	0.3174238092794097	0.5716273865401686	0.1357666114893007
Ce	0.5695737576227561	0.7370220635634706	0.1341994549591767
Ce	0.0701476151266783	0.7358156531966842	0.1363243184266147
Ce	0.0694792662019655	0.4048399080045133	0.1354190045725545
O	0.8118050505362239	0.2320881381818570	0.3784801976915113
O	0.6526543843667126	0.2327815449080468	0.3205399685007758
O	0.4085288731389640	0.3944515483456982	0.3166161996633592
O	0.6577204366528978	0.5744272895286627	0.3211963417534063
O	0.5666224918648491	0.7564525469055839	0.3835726295448467
O	0.8122335858041652	0.5673409898393283	0.3774360306726587
O	0.8136098085857116	0.9009418354839231	0.3777535627688796
O	0.5747956282431430	0.3940706596291917	0.3774906612602145
O	0.8954698049743059	0.7355323372257199	0.3221671711435632
O	0.8978872061488002	0.3966710350566987	0.3223771212638655
O	0.0620020656251710	0.0617212662707366	0.3775759358367827
O	0.1437719421780501	0.2519599300774608	0.3377186900175297
O	0.3108329611881650	0.2449053328755751	0.3839605039809182
O	0.6506472701043667	0.9034205169884695	0.3217794844774177

O 0.4012353305598153 0.0790531891182226 0.3230838211034905
 O 0.4103268664078804 0.7362418451177635 0.3253655090041012
 O 0.5699174634574283 0.0735213030924932 0.3778176109841650
 O 0.8970920271932977 0.0673293485856393 0.3228062977624039
 O 0.1309669998094661 0.5861073411953245 0.3234259977283049
 O 0.1464564002211516 0.9044107215748269 0.3224280825560850
 O 0.3212141284249307 0.9188524096627456 0.3789819814054560
 O 0.0680407885853528 0.0693890371663051 0.2138030967473322
 O 0.5688306755250967 0.0684102188049728 0.2135104388188304
 O 0.7357275105490542 0.0685202741967319 0.2688049060583274
 O 0.2286215156017201 0.0792050942262798 0.2727806741449939
 O 0.8209281920622937 0.2354907060807278 0.2146387851921722
 O 0.9902732065333772 0.2330495166677614 0.2714358193429487
 O 0.5690846000777863 0.7377852412597671 0.2132014415527367
 O 0.8183678263956417 0.5695341446028754 0.2134809048714451
 O 0.8192724846572811 0.9034738474735142 0.2141858323089063
 O 0.5698488249666108 0.4031589015381034 0.2119772878580207
 O 0.4882557261788346 0.5719982318601790 0.2666156906335576
 O 0.7352046485507910 0.4028710142723613 0.2686904084126667
 O 0.7343129828860210 0.7376264289695534 0.2686198047173092
 O 0.4888976664454088 0.2340432681136768 0.2670010563799170
 O 0.9841558757139302 0.9036414210617020 0.2704863037921544
 O 0.9789154603296306 0.5661699104538979 0.2693675419436863
 O 0.0674142458342750 0.4030453561632013 0.2148793154892134
 O 0.3182742721703662 0.2359465799509391 0.2130689265743328
 O 0.3196507507750675 0.5697973634281938 0.2143883573474480
 O 0.2339166834978862 0.4027567643483262 0.2676105050707889
 O 0.4848994043852280 0.9051422974189094 0.2698667980642364
 O 0.3174834657158340 0.9084107644110034 0.2152962816946702
 O 0.0685645633290509 0.7373752918916384 0.2166995923845800
 O 0.2393960995542141 0.7367113559276499 0.2682565218203886
 O 0.9883231168672951 0.9032199487805642 0.1077878321802966
 O 0.4867717852454109 0.9032044407533413 0.1076235641420885
 O 0.6539729943646497 0.9032623284145964 0.1614728478529310
 O 0.1529503986204326 0.9033141809481865 0.1630039757712795
 O 0.7384533107577648 0.0698674890660476 0.1073216633154522
 O 0.9037386733858973 0.0698440034002988 0.1620712524423600
 O 0.4870215346225001 0.5719285739121271 0.1079039864070674
 O 0.7378566726715151 0.4034298459574670 0.1068383289190196
 O 0.7387424064115711 0.7370627058394850 0.1073178421080069
 O 0.4880217857133144 0.2363481804457584 0.1069577297915810
 O 0.4033434690877076 0.4027940172259045 0.1605665091119437
 O 0.6547994323446715 0.2362566267805149 0.1614377695924342
 O 0.6541402318711270 0.5705743830865033 0.1609495003173365
 O 0.4033195036743870 0.0707226453101512 0.1618380309543987
 O 0.9043027984801059 0.7365087047999658 0.1626052080478422

O	0.9032120133763777	0.4031461217626857	0.1620064581323729
O	0.9875746122185247	0.2365289351353999	0.1075298599973922
O	0.2370719551467958	0.0701019865060121	0.1075216605460231
O	0.2356404531786817	0.4043486934114403	0.1082427721228037
O	0.1525165432241932	0.2363351756534934	0.1622678459899718
O	0.4022692453374568	0.7388482276153685	0.1630816969300736
O	0.2367281500145925	0.7367438389488626	0.1084937686633128
O	0.9884623386769050	0.5703485047570224	0.1077950295215700
O	0.1522010399912778	0.5710127817753337	0.1639546601344597
O	0.2162456473268573	0.3978561515882569	0.4572396499860609
O	0.4322543918708052	0.5919998107566342	0.4469265106565385
O	0.2126726128337075	0.6695567164654720	0.4462740575427193
O	0.3081370051335107	0.5585353125518233	0.3807534444847972
O	0.0418585383138986	0.7308102380475867	0.3853449099641304
O	0.0308697983073101	0.3925925179122681	0.3925651809253647

H^* on $Pd_7O_9/Ce_{33}O_{66}(111)$

Total DFT Energy: -886.38979639 eV

Unit Cell (Å):

13.3200000000000003	0.0000000000000000	0.0000000000000000
0.0000000000000000	11.3610000000000007	0.0000000000000000
0.0000000000000000	0.0000000000000000	30.0000000000000000

Fractional Coordinates:

H	0.0049102045936630	0.3963964233946546	0.4353192473952271
Pd	0.5263029687434472	0.4974669251733293	0.3262280035944922
Pd	0.2747230111158400	0.8164366538578219	0.3271253174046962
Pd	0.2728433133087655	0.3368623168222211	0.3274345045354399
Pd	0.4301366552629373	0.6571617383995647	0.3858453932069630
Pd	0.1779934688557201	0.6482410060565008	0.3834748411690928
Pd	0.2952956770152522	0.5364302550535692	0.4576892662839542
Pd	0.1789148389346529	0.3238505017603320	0.4060148042004039
Ce	0.9732604219166948	0.2241150264180760	0.3512613058348608
Ce	0.7377925529642223	0.4001702861269027	0.3490351254674645
Ce	0.4931707509925963	0.2344827338924858	0.3486177235828446
Ce	0.7275658074194478	0.7383475659563645	0.3515519023169345
Ce	0.9804530944729556	0.8963032009627725	0.3511646098684268
Ce	0.9731708258269617	0.5734311274221283	0.3497764072966617
Ce	0.7330907568513676	0.0657845662446621	0.3488829510553469
Ce	0.4871809078559397	0.9101511829775988	0.3507725506019920
Ce	0.2300484739948160	0.0702495067014740	0.3516750223641618
Ce	0.4022706495462266	0.0712704937715575	0.2427377037141414
Ce	0.9019343809752156	0.0674151985792481	0.2416685663227185
Ce	0.4046831977161609	0.4023242088423973	0.2379709883150891
Ce	0.6549824192287534	0.5706020787774919	0.2396293513649610
Ce	0.6538625284054306	0.2354950838014507	0.2405622550594942
Ce	0.8997700781952027	0.7363266210396401	0.2418001173209582

Ce	0.9023654391977394	0.4012323548900065	0.2419462823364917
Ce	0.1507105363722165	0.2374375846671446	0.2388568139644902
Ce	0.4040535810654375	0.7375424265281744	0.2421389850766851
Ce	0.6519277539906219	0.9030479482655517	0.2409379866170623
Ce	0.1500591132283057	0.9055980000354449	0.2424060711221325
Ce	0.1494423480204871	0.5721005397217245	0.2466099827906360
Ce	0.3193817467419655	0.9042664177395062	0.1357105977957793
Ce	0.8198175677654859	0.9029899772474895	0.1339953030883953
Ce	0.3199498596946833	0.2358000234323060	0.1337706459006247
Ce	0.5712783083711410	0.4030249353686806	0.1331575122469949
Ce	0.5700052518472795	0.0690556844940714	0.1343366640129536
Ce	0.8204139082414060	0.5703406956718515	0.1339769273788176
Ce	0.8206634814766498	0.2362172854549165	0.1339905198163461
Ce	0.0688954088859875	0.0686656971109851	0.1341801668829115
Ce	0.3176376742139786	0.5713677956667828	0.1360056757575002
Ce	0.5690433225765351	0.7367251486696584	0.1339534680476941
Ce	0.0704157384324639	0.7356013071822498	0.1361492042336660
Ce	0.0694992687411710	0.4045999479338415	0.1352103949461959
O	0.8146020986605118	0.2310385005278623	0.3781930749522087
O	0.6543556890320751	0.2319729722019612	0.3202636566859339
O	0.4097775547018234	0.3920554616775695	0.3159857303296556
O	0.6592015539749624	0.5738774822128619	0.3210865823594097
O	0.5666132722535636	0.7564179682942410	0.3834497107901045
O	0.8140147828332237	0.5672806900934493	0.3775098107852228
O	0.8125007753210012	0.9011755353693259	0.3778312867158340
O	0.5781261307075782	0.3930946146254745	0.3772682899084503
O	0.8942158253416593	0.7365950523206656	0.3216769281683634
O	0.9043944562033726	0.3963678891299655	0.3237338100537190
O	0.0590770938016612	0.0632175495658050	0.3771356806819163
O	0.1413018215244755	0.2594589860732543	0.3409073480507350
O	0.3126885433995409	0.2447203548739129	0.3839772816930137
O	0.6501618124843492	0.9032499104435653	0.3216120195364435
O	0.4014209814615204	0.0798088060378918	0.3228359467035461
O	0.4096748297646207	0.7379276735580381	0.3251298778815900
O	0.5702533463401613	0.0735137845085496	0.3776752256825870
O	0.8960952208690512	0.0648819583197771	0.3219767157260974
O	0.1283431975839459	0.5852547091915347	0.3232023424235576
O	0.1453549841152461	0.9065404263352130	0.3225559924463227
O	0.3200349471944072	0.9218455968976982	0.3787173339095252
O	0.0681355870183864	0.0694434683978312	0.2134214206345580
O	0.5692433449929258	0.0680853457319776	0.2128922059767742
O	0.7357419264143801	0.0676924990058454	0.2682259030011669
O	0.2277545849182334	0.0806720382554275	0.2730334265927101
O	0.8214691986657326	0.2352998358284832	0.2143184199373039
O	0.9908904670296219	0.2309478854233403	0.2716810937919976
O	0.5688240278297062	0.7377346329088916	0.2129583144121720

O 0.8179143401291095 0.5690247842633368 0.2129931848435181
 O 0.8189159664272713 0.9032106808223651 0.2133884698372559
 O 0.5712043489617984 0.4029611590476705 0.2119477024834793
 O 0.4888635261464491 0.5723500962526482 0.2664963730338645
 O 0.7384689087435811 0.4023751359437613 0.2691145339198598
 O 0.7339009335010978 0.7379217599419585 0.2681584604110595
 O 0.4904760070868778 0.2327624165975116 0.2664444989853110
 O 0.9838453293271661 0.9031226446524809 0.2698190859881747
 O 0.9774565504242441 0.5676420326495945 0.2687854558618101
 O 0.0668978651633081 0.4029040275898705 0.2148366409198230
 O 0.3186250750017208 0.2357099684503709 0.2130232086570728
 O 0.3196466629688127 0.5698959546054190 0.2146603222421013
 O 0.2330072606689163 0.4033080238982356 0.2673541629122131
 O 0.4854638104272156 0.9057664585545695 0.2696837870973782
 O 0.3174331105797327 0.9087314591350614 0.2154107196031524
 O 0.0683917987182427 0.7374594669492218 0.2162287160468585
 O 0.2385490403582303 0.7370260997041216 0.2683584649096276
 O 0.9886343943716770 0.9029394977647426 0.1075539873233478
 O 0.4862881486421990 0.9028965527010810 0.1073871956845425
 O 0.6538490823005672 0.9028644412033320 0.1609184835621713
 O 0.1530892151260431 0.9031702302860946 0.1629752279324606
 O 0.7383066479388816 0.0696366414698992 0.1067373802343585
 O 0.9034168586730242 0.0695708037872517 0.1615666596294682
 O 0.4864518363309300 0.5716217910300692 0.1077400953775417
 O 0.7378174861960813 0.4032877913055153 0.1064866513305408
 O 0.7386831233476113 0.7366847869204979 0.1067291083960119
 O 0.4879059226672429 0.2365785387573877 0.1068265153207921
 O 0.4037131330863954 0.4027700847153992 0.1608027290097958
 O 0.6551088244513471 0.2359536221085880 0.1610746942777790
 O 0.6538095489170270 0.5707790341842642 0.1606011884153724
 O 0.4034510807834272 0.0706432242181880 0.1616520184762016
 O 0.9039419716113211 0.7362893071180567 0.1620337018481175
 O 0.9030538904580143 0.4028746726836240 0.1618417408642550
 O 0.9877951871608134 0.2365151604757626 0.1074077390532070
 O 0.2368797311514205 0.0702073104716487 0.1075272063450032
 O 0.2359893330564698 0.4042416138448136 0.1082391072627094
 O 0.1527765460566968 0.2366252904511302 0.1621832399355546
 O 0.4021102902440371 0.7387818824773243 0.1631082947954115
 O 0.2365637819741291 0.7363840666571599 0.1085639364389078
 O 0.9886693492553043 0.5698474718348940 0.1076352661983628
 O 0.1522442656025547 0.5704597387995762 0.1639914756826234
 O 0.2263343375784323 0.3914004729605938 0.4613437925360673
 O 0.4270687264674063 0.5934068377233462 0.4471562947217024
 O 0.2091939848285409 0.6682565566848321 0.4469675800917677
 O 0.3064247644479969 0.5626266313717692 0.3779922889870396
 O 0.0395213925769170 0.7327163768852063 0.3853595031509047

O 0.0302912625701536 0.3901973001266860 0.4049265192427906

Pd₇O₁₀/Ce₃₃O₆₆(111)

Total DFT Energy: -885.85977944 eV

Unit Cell (Å):

13.3200000000000003	0.0000000000000000	0.0000000000000000
0.0000000000000000	11.3610000000000007	0.0000000000000000
0.0000000000000000	0.0000000000000000	30.0000000000000000

Fractional Coordinates:

Pd	0.5239712559774290	0.4911982256237900	0.3262694124480785
Pd	0.2723540788113665	0.8070875306703114	0.3273901093571914
Pd	0.2624011600702978	0.3313967094056066	0.3265715697092175
Pd	0.4290705761983596	0.6480418725747794	0.3868720156413988
Pd	0.1819044857830208	0.6553079970701176	0.3830070180501812
Pd	0.3009020654401415	0.5761809478298401	0.4678228730743518
Pd	0.2025218174611233	0.3941262383749872	0.4066925905937198
Ce	0.9824142074507454	0.2232184966512124	0.3538241690640117
Ce	0.7400649986195791	0.3956089141641964	0.3498250832833690
Ce	0.4935342908744423	0.2307833867048576	0.3482840924360297
Ce	0.7280330836606156	0.7341192787206943	0.3507446799991956
Ce	0.9807248588932195	0.8965297792098124	0.3497756602373557
Ce	0.9742851787102440	0.5735602410425379	0.3504854062254519
Ce	0.7349579328974337	0.0636799597779767	0.3492755591410029
Ce	0.4860027411738769	0.9050552899725006	0.3507971566670205
Ce	0.2268295837672670	0.0665080082525793	0.3502533696383606
Ce	0.3998305844003905	0.0662049550349728	0.2420316563504486
Ce	0.8996888589848727	0.0653131027658723	0.2414937242911522
Ce	0.4034085244697646	0.3976931352475660	0.2383534021477019
Ce	0.6545200813772561	0.5659988862081744	0.2401241236266680
Ce	0.6525243836525977	0.2312690627199528	0.2410344588965717
Ce	0.8992615797073137	0.7329715304690511	0.2416031687823393
Ce	0.9017665723163014	0.3973816158082544	0.2419575164733820
Ce	0.1472125553254828	0.2307569360048983	0.2396884166839690
Ce	0.4031044922674963	0.7313696099729680	0.2407875716855627
Ce	0.6505160488885571	0.8987715003805616	0.2414843793106472
Ce	0.1489507478638608	0.9010811208423476	0.2401420479776766
Ce	0.1483785959337509	0.5681802591076633	0.2453234102138275
Ce	0.3184514013142673	0.8995598130368968	0.1347520791277596
Ce	0.8178408673678592	0.8992698821814950	0.1339373208576723
Ce	0.3182763398774965	0.2313525404486626	0.1338489786644344
Ce	0.5693767090138195	0.3993577647053824	0.1332083911934051
Ce	0.5686297818415481	0.0655811019561601	0.1342761825537900
Ce	0.8187291032740526	0.5660818443541396	0.1339496613682403
Ce	0.8181995339671042	0.2321343157786556	0.1339348183095762
Ce	0.0669487327272516	0.0657933983978127	0.1331710629671806
Ce	0.3167949770390929	0.5665193616993573	0.1351907728434323

Ce	0.5685110060646816	0.7329272053328476	0.1336800847262472
Ce	0.0685607775223359	0.7316027450973063	0.1351059328091948
Ce	0.0682919859145563	0.4003430190842610	0.1352021905035469
O	0.8127314765265305	0.2286587721666843	0.3779725163165099
O	0.6529610161643415	0.2283353248598455	0.3202945227517464
O	0.4037683071353163	0.3875372234481224	0.3189376582377835
O	0.6572504784131906	0.5682109547878513	0.3212074649167384
O	0.5612344957477258	0.7499419192695903	0.3829604680815769
O	0.8134048402429922	0.5674786976953434	0.3773281456950699
O	0.8111090364693220	0.8975807529879964	0.3771845970666108
O	0.5774685689385982	0.3890059898937137	0.3773751643126098
O	0.8946965282108363	0.7337342681525223	0.3215012795785828
O	0.9100245069672289	0.3970061214218088	0.3271175713594149
O	0.0665885365467446	0.0551127888446354	0.3765960108524970
O	0.1406065486998909	0.2297062053398932	0.3197462654264351
O	0.2942057030237166	0.2520456130520876	0.3850108449415048
O	0.6485666292053694	0.8979036278846653	0.3217219657761578
O	0.3973900251685991	0.0781795329117028	0.3243619788240407
O	0.4064643184746037	0.7273633358142457	0.3258418744293114
O	0.5689307474206168	0.0686120983812455	0.3771460169672388
O	0.8989749835570794	0.0636813907648747	0.3227593148666504
O	0.1294587107538496	0.5894910109037360	0.3228303137923385
O	0.1459706323824878	0.9008610547997612	0.3204060199631871
O	0.3177256816959798	0.9123065025519962	0.3779801045380491
O	0.0666774981609392	0.0650353811853185	0.2128637381243106
O	0.5680139751459655	0.0632697444171437	0.2136724675124300
O	0.7349660478327572	0.0632737791682859	0.2690347373156055
O	0.2332773389175852	0.0658954107550876	0.2691247765825946
O	0.8161308724178068	0.2314753317264741	0.2132163939060242
O	0.9807705032990307	0.2296841863110464	0.2680983723055683
O	0.5680177664945246	0.7335910831900434	0.2134097834433696
O	0.8164922669984798	0.5657923622901901	0.2131415282238724
O	0.8161946660103604	0.8988224587657794	0.2134082268657218
O	0.5711014690976314	0.3986027805214999	0.2126491231917240
O	0.4864971801500775	0.5661308848309589	0.2664685332906248
O	0.7401029834794405	0.3978326710838739	0.2703250870163716
O	0.7332576970127390	0.7331224814654980	0.2687505998930656
O	0.4879979386803661	0.2292493592084053	0.2671449895909838
O	0.9828051953184458	0.8997317026460345	0.2686179930426465
O	0.9780136832433403	0.5644058050098955	0.2686726312326680
O	0.0667111389270806	0.3992602489321052	0.2146154347555485
O	0.3171140597464912	0.2318912736270035	0.2131604168226514
O	0.3183029983985386	0.5656464075706128	0.2137844333677132
O	0.2337290667705690	0.4002603809032721	0.2662019908116618
O	0.4807180680238221	0.9003154898280262	0.2707334392048870
O	0.3176097614977561	0.9000556639162390	0.2135233031724707

O	0.0673758897381537	0.7325356954271911	0.2151271716351792
O	0.2386159320116213	0.7324352449735109	0.2675359277203404
O	0.9860409775311429	0.8994546353378112	0.1067823769799450
O	0.4853247241979575	0.8992587676671396	0.1069692020394413
O	0.6520460602795772	0.8987006036664482	0.1609572551808564
O	0.1513488311219408	0.8989016962413322	0.1615549883799046
O	0.7361189488294666	0.0656268748083220	0.1066138031096136
O	0.9015701228240379	0.0657146269008931	0.1612436039247853
O	0.4859285314870986	0.5664581816211709	0.1070993454746656
O	0.7363137012369404	0.3993642781202491	0.1066679431404578
O	0.7367191772671077	0.7325776631699080	0.1065926977494998
O	0.4855452450023386	0.2325077161244354	0.1065461986027174
O	0.4030663501415560	0.3985653095300683	0.1611296169971153
O	0.6517452498310659	0.2314566230710174	0.1608381330645665
O	0.6523273169613313	0.5667226615214620	0.1608512801181617
O	0.4021150479609115	0.0661366671157235	0.1617399535659893
O	0.9023982837916236	0.7326116524591126	0.1617721361029541
O	0.9017807748707222	0.3991047694372399	0.1619835416237800
O	0.9855990622617263	0.2327704058784960	0.1068553424574513
O	0.2360949756967128	0.0657554565760049	0.1069693746836822
O	0.2352062936932875	0.4001633839950291	0.1080238431768850
O	0.1509112114420887	0.2320627749944283	0.1613660792107357
O	0.4023764558467084	0.7329113769487221	0.1620885818128985
O	0.2356358603217284	0.7317088158009957	0.1078963049446034
O	0.9866907866189680	0.5658306766829160	0.1073318509592606
O	0.1509847113569195	0.5664132598923182	0.1632732112686541
O	0.2354026668639003	0.4317206524675996	0.4705554040547467
O	0.4320416060329848	0.6089744274564612	0.4499485997116179
O	0.2216176888416779	0.7015732085719971	0.4449293903914482
O	0.3027277931788431	0.5507384171307911	0.3767008103772150
O	0.0456856022486441	0.7425165285244574	0.3841127503996717
O	0.1029714508125745	0.5252421350465928	0.4096730372436870
O	0.0755221654290011	0.3049691568739521	0.4030600276005351

*H** on $Pd_7O_{10}/Ce_{33}O_{66}(111)$

Total DFT Energy: -891.15334962 eV

Unit Cell (Å):

13.3200000000000003	0.0000000000000000	0.0000000000000000
0.0000000000000000	11.3610000000000007	0.0000000000000000
0.0000000000000000	0.0000000000000000	30.0000000000000000

Fractional Coordinates:

H	0.0421320890871598	0.7433482354907083	0.4290985725170903
Pd	0.5224678178009069	0.4910650284641827	0.3271292646426479
Pd	0.2726740955954151	0.8089442749943537	0.3276623937320750
Pd	0.2565341106036296	0.3306948693412637	0.3299845733498302
Pd	0.4391405909250823	0.6605569451852111	0.3843428236053190

Pd	0.1895519032898102	0.6578880032482973	0.3880527498150785
Pd	0.3240001809950499	0.5702444065131003	0.4657310917022374
Pd	0.1738987150905488	0.3866336615622430	0.4155424865037834
Ce	0.9750846271216854	0.2226307507439643	0.3478540821256205
Ce	0.7371893826953798	0.3955716993432585	0.3484006421923905
Ce	0.4935058611570179	0.2319429658263668	0.3470082394361158
Ce	0.7319937581726529	0.7326752175995500	0.3503576406070242
Ce	0.9769678855993206	0.8984725604173948	0.3478466569374034
Ce	0.9802495895321699	0.5528867479760738	0.3545999674676501
Ce	0.7326352654952721	0.0636420773819086	0.3473909716673991
Ce	0.4873832423622649	0.9101288533666405	0.3498013109876005
Ce	0.2244583150462477	0.0675427244448653	0.3509684400328654
Ce	0.3991620547219735	0.0663292064749655	0.2418914337357619
Ce	0.8988369703507022	0.0635335647003129	0.2390501268796386
Ce	0.4011168671432325	0.3979835918249725	0.2397153384076609
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